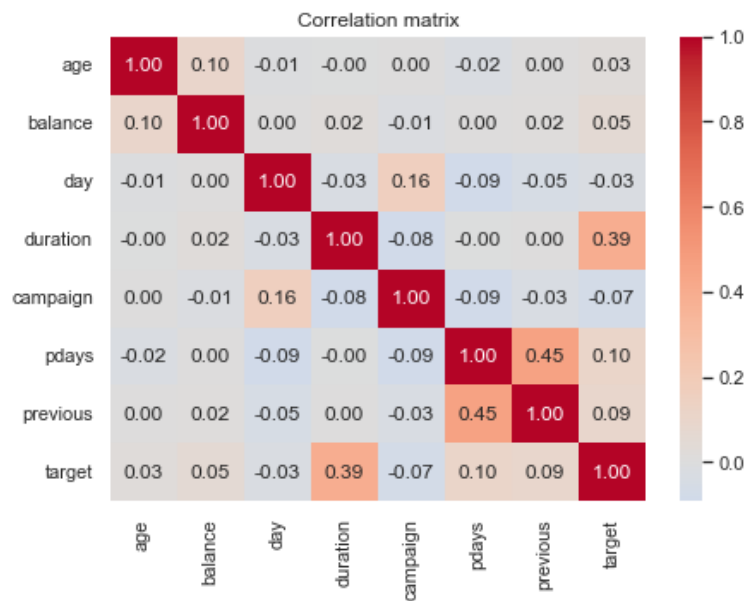
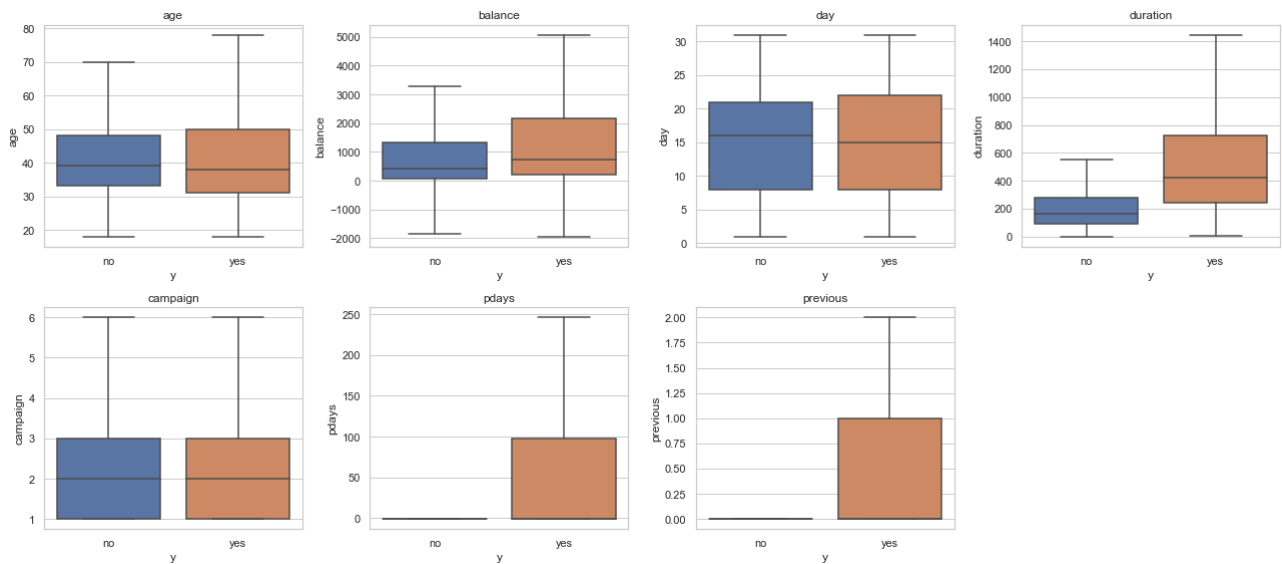


Bank Marketing: Term Deposit Subscription

Visual summary of the Jupyter notebook "Bank Marketing.ipynb" (UCI Bank Marketing, bank-full.csv, 45,211 customers, 11.7% subscribed). Random seed 42, stratified 80/20 split, *duration* excluded to avoid leakage.

1. Exploratory analysis





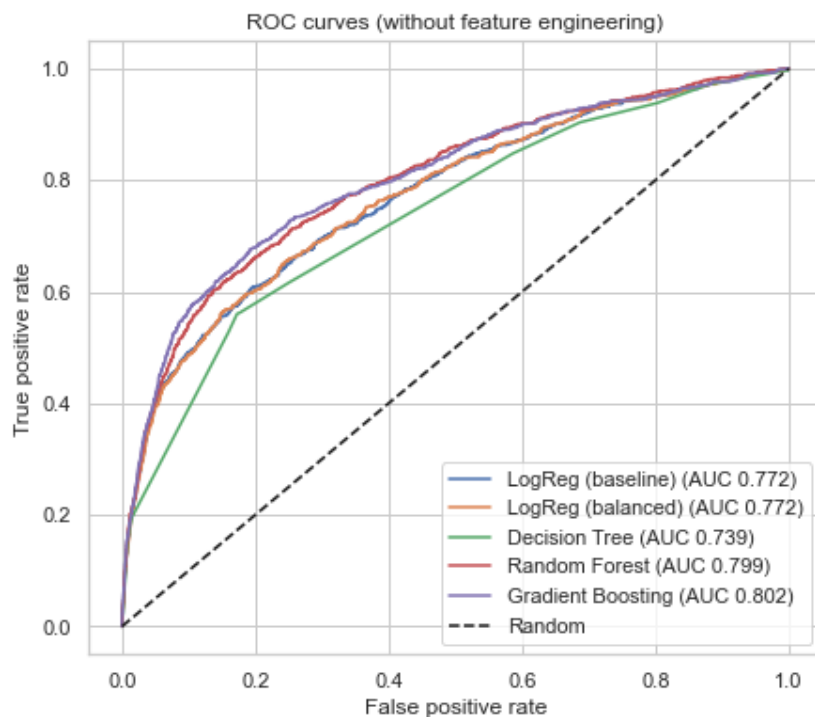
Subscription rate is highest after a previous success (64.7%), for under-25s (24.0%) and over-65s (42.6%), students and retired customers, and customers with no loans.

2. Baseline and model comparison

Test-set metrics of all models (executed output of the notebook; "+ FE" = with engineered features).

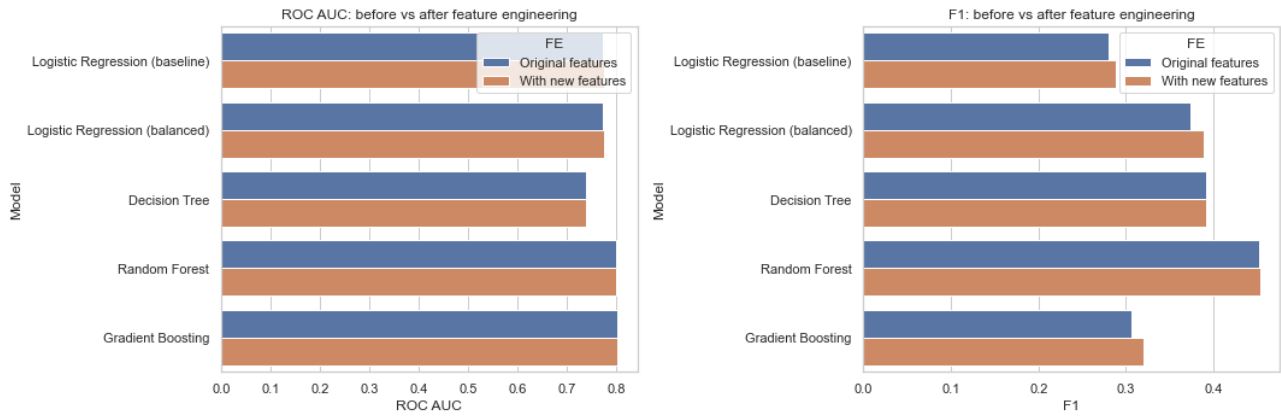
	Accuracy	Precision	Recall	F1 \
Model				
Logistic Regression (baseline)	0.893	0.664	0.178	0.280
Logistic Regression (balanced)	0.755	0.267	0.624	0.374
Decision Tree	0.796	0.301	0.560	0.391
Random Forest	0.828	0.361	0.604	0.452
Gradient Boosting	0.894	0.659	0.199	0.306
Logistic Regression (baseline) + FE	0.893	0.640	0.186	0.288
Logistic Regression (balanced) + FE	0.764	0.279	0.642	0.389
Decision Tree + FE	0.810	0.313	0.522	0.391
Random Forest + FE	0.834	0.368	0.590	0.453
Gradient Boosting + FE	0.895	0.666	0.211	0.320

	ROC AUC
Model	
Logistic Regression (baseline)	0.772
Logistic Regression (balanced)	0.772
Decision Tree	0.739
Random Forest	0.799
Gradient Boosting	0.802
Logistic Regression (baseline) + FE	0.775
Logistic Regression (balanced) + FE	0.775
Decision Tree + FE	0.738
Random Forest + FE	0.799
Gradient Boosting + FE	0.802



Baseline Logistic Regression: accuracy 0.893 but recall for "yes" only 0.178. Class-weighted models and the Random Forest find 56-64% of subscribers; Gradient Boosting and Random Forest have the best ROC AUC (0.802 and 0.799).

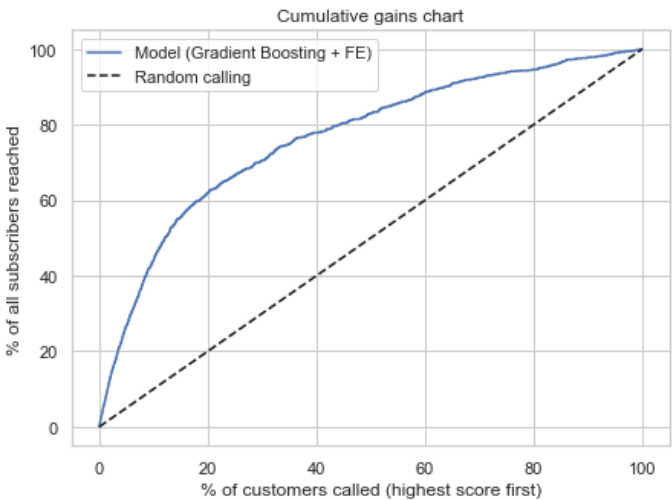
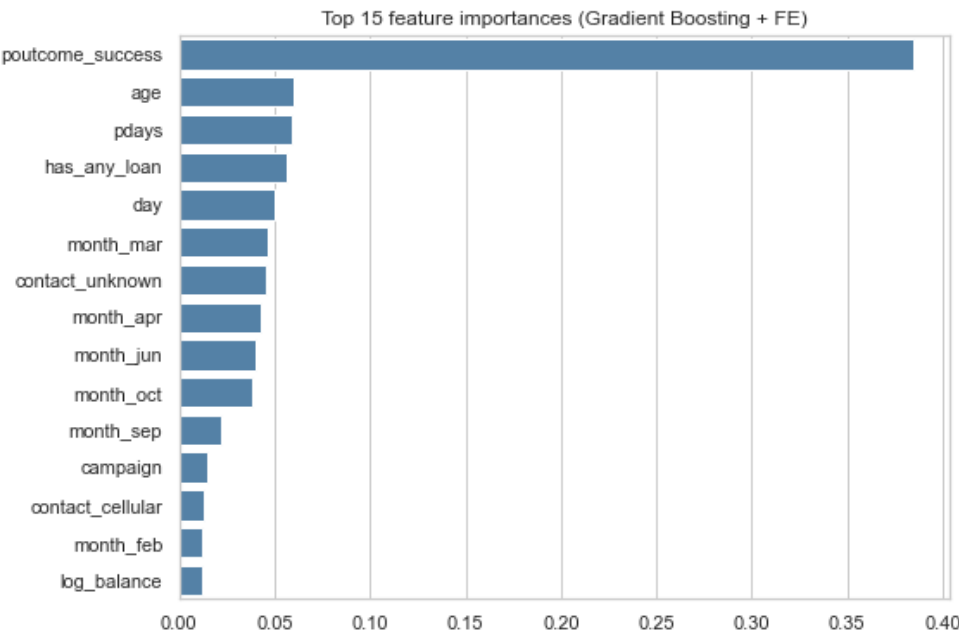
3. Feature engineering



New features: `contacted_before`, `has_any_loan`, `balance_positive`, `log_balance`, `age_group`, `season`, `campaign_high`. Effect: small (ROC AUC +0.003 for Logistic Regression; F1 +0.014 for Gradient Boosting; no change for Random Forest).

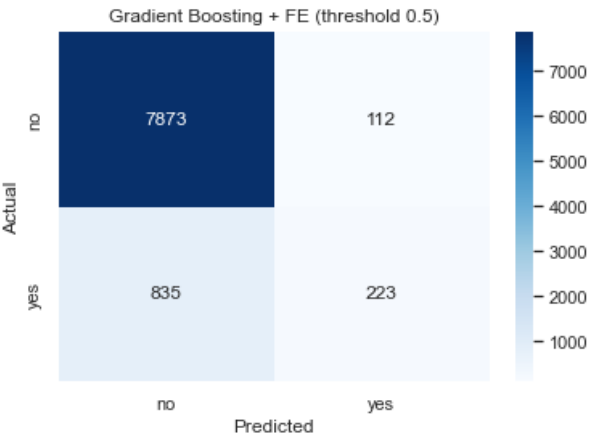
Leakage check: including *duration* raises ROC AUC from 0.802 to 0.925, but it is only known after the call, so it is excluded.

4. Interpretation



Business use: gains table (Gradient Boosting + FE)

Customers called (%)	Subscribers reached (%)	Success rate among called (%)	Lift vs random
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Final conclusions (business view)

What drives subscriptions

- Only 11.7% of customers subscribed, so the data is imbalanced and accuracy alone is misleading (always saying "no" gives 88.3%).
- The strongest signal is the **previous campaign outcome**: customers with a previous success subscribed 64.7% of the time versus 9.2% for customers with no previous contact. It is also by far the most important feature in Gradient Boosting (importance 0.385, the next one is 0.060).
- Other helpful signals: **age** (under 25: 24.0%, over 65: 42.6%, versus about 9-12% for ages 26-55), **job** (students 28.7%, retired 22.8%, blue-collar 7.3%), **no housing or personal loan** (18.2% versus 7.7% for the new `has\_any\_loan` flag), **known contact type** (`unknown` contact: 4.1% versus 14.9% cellular), and **fewer calls** (more than 3 calls: 7.4% versus 12.9%).
- Timing matters (March, September, October, December convert best), but these months have few contacts, so we should test this before changing the calendar.

Model comparison (test set, 9,043 customers, `duration` excluded)

Model	Precision	Recall	F1	ROC AUC
Logistic Regression (baseline)	0.664	0.178	0.280	0.772
Logistic Regression (balanced)	0.267	0.624	0.374	0.772
Decision Tree	0.301	0.560	0.391	0.739
Random Forest	0.361	0.604	0.452	0.799
Gradient Boosting	0.659	0.199	0.306	0.802
Gradient Boosting + new features	0.666	0.211	0.320	0.802

- **Gradient Boosting and Random Forest rank customers best** (ROC AUC about 0.80, versus 0.77 for the Logistic Regression baseline and 0.74 for the Decision Tree).
- The baseline Logistic Regression and Gradient Boosting have high precision but find only 18-21% of subscribers at the default 0.5 threshold. The "balanced" models and the Random Forest find about 56-64% of subscribers, but with more false alarms. **Random Forest has the best F1 (0.45)**. The choice between these depends on the cost of a call versus the value of a subscription.
- **Feature engineering helped only a little** (ROC AUC +0.003 for Logistic Regression, F1 +0.014 for Gradient Boosting, no change for Random Forest). The tree models can already find these patterns from the raw variables.
- **Leakage check**: with `duration` included, ROC AUC jumps from 0.802 to 0.925, but that variable is only known after the call, so it cannot be used to plan a campaign.

How the bank can use this

- Score every customer before the campaign and call them in order of score. With Gradient Boosting, calling the **top 10%** reaches 44.1% of all subscribers (51.7% success rate, 4.4 times better than random); the **top 20%** reaches 62.0% (3.1 times better); the top 50% reaches 83.0%. This saves many calls.
- Prioritise customers with a previous successful campaign, students, retired and older customers, and customers without loans; avoid repeated calls (more than 3) to the same person.
- Use the Random Forest or balanced models when the goal is to find as many subscribers as possible, and Gradient Boosting when the budget is tight and precision matters.

Limitations

- Data comes from one bank and one past period; results may change for new campaigns, so a small pilot test is recommended.
- Results are from a single train/test split with default or lightly set hyperparameters; no tuning or cross-validation was done, so small differences (for example between the two ROC AUC values 0.802 and 0.799) are not

conclusive.

- Patterns are correlations, not proof that a characteristic *causes* subscription.
- Next steps: cross-validation, hyperparameter tuning, choosing the decision threshold from real call costs and deposit value.